

Git source code storage and version control

Michael Hart-Davis¹

¹ Deutsches Geodätisches Forschungsinstitut, Technische Universität München, Germany

Date: 27 April 2026

Git

Installing Git:

<https://git-scm.com/downloads>
[download it based on your OS]

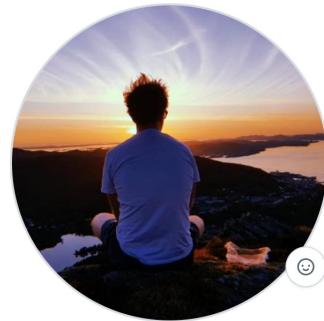
Sign in and get it working!
TUM has a license with GITLAB:
https://gitlab.lrz.de/users/sign_in

Why Git?

- Version control of your codes for the numerical modelling assignment.
- This not only helps you, but it can also be useful to help me help you!
- As you are working in groups, this allows you to collaborate nicely and keep track on who does what!
- In some cases, git can also be used to help check the workability of your code.
- Its not required that you do, but in future work and collaboration, git is one of the most used platforms for code sharing.
- I use Git regularly, and it is an incredible resource for collaborating. I will show you an example of oceanparcels:

<https://github.com/Parcels-code/Parcels>

A note on GitHub



hart-davis

PhD in Satellite Oceanography. Research Associate at DGF-TUM. Work with Fortran, Python, Matlab, Latex and BASH.

Edit profile

5 followers · 8 following

Technical University of Munich

mhartd@gmail.com

Achievements



Organizations



Popular repositories

[Customize your pins](#)

Particle-Tracking-Project

Public

Project for the CSIR

Python 1

codes

Public

my codes

Jupyter Notebook 1

parcels

Public

Forked from [OceanParcels/Parcels](#)

Main code for Parcels (Probably A Really Computationally Efficient Lagrangian Simulator)

Jupyter Notebook

SinglePaged

Public

Forked from [t413/SinglePaged](#)

SinglePaged - Simple Jekyll template

CSS

Flaskex

Public

Forked from [anfederico/flaskex](#)

Simple flask example for quick prototypes and small applications

CSS

bulma

Public

Forked from [jgthms/bulma](#)

Modern CSS framework based on Flexbox

CSS

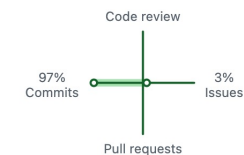
141 contributions in the last year

Contribution settings



Activity overview

Contributed to [hart-davis/eot20_tides](#), [backeb/mike-bjorn-collab](#)

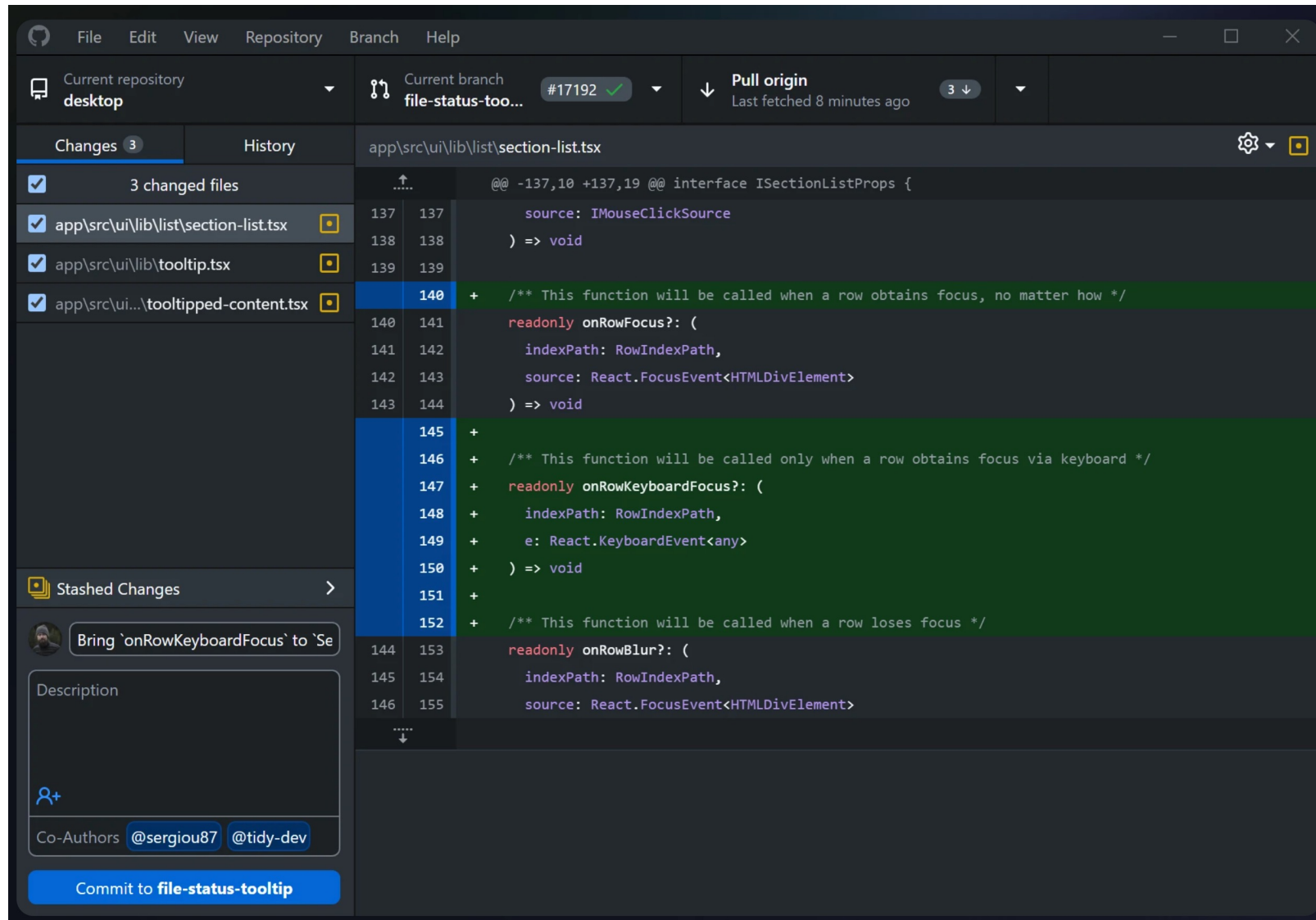


Contribution activity



Eg. <https://github.com/hart-davis/ArcTiCA>

The GitHub App



Gitlab use

```
git clone https://gitlab.lrz.de/ge24man/advanced-numerical-modelling.git
```

```
# you need an account on gitlab.lrz.de (your TUM login)
```

```
git branch HARTDAVIS
```

```
git checkout HARTDAVIS
```

```
git push --set-upstream origin HARTDAVIS
```

```
# this creates your own unique branch
```

```
git push origin mikehd
```

```
# pushes the branch to the gitlab
```

Now my branch is online

A

Advanced Numerical Modelling

 main ▾

advanced-numerical-modelling /

 ▾

Select Git revision

 Search by Git revision

Branches 2

 main default protected

mikehd

minutes ago

Last commit

Initial commit

 README.md

Now what do we do

```
# make a change to the README.md file, add your name or something.  
# then add these changes
```

```
git add 'README.md'
```

```
# write a message documenting what you changed  
git commit -m 'changing the README of my branch'
```

```
# push your git to your branch!  
git push
```


Git can do so much more!

- I find I use the community templates quite often to set up projects (e.g.):

<https://github.com/jbusecke/cookiecutter-science-project>

- Will demonstrate an example of a well set up project within GIT! (drifter paper project on private page) or pyTMD:

<https://github.com/pyTMD/pyTMD>

- Will demonstrate an example of a CV or webpage (my embarrassing MSc online CV).

Lets try live example of me using it!